Mechanics Year 1 · Chapter 10

Forces and Motion

Force diagrams and equilibrium

Newton's laws

Connected particles and pulleys

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STARTER 2

1. Calculate $(2\mathbf{i} + \mathbf{j}) + (3\mathbf{i} - 4\mathbf{j})$ and $(-\mathbf{i} + 3\mathbf{j}) - (3\mathbf{i} - \mathbf{j})$.
2. A right triangle has perpendicular sides 12 cm and 15 cm. Find its hypotenuse and the angle opposite the 12 cm side to 1 d.p.
3. A car accelerates from rest at $1.5\ \text{m/s}^2$ for 12 s. Find its speed. It then decelerates at $1\ \text{m/s}^2$: find its stopping distance.

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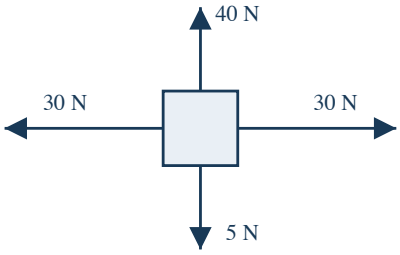
Force diagrams

Weight $W = mg$ acts vertically downwards. Normal reaction R is perpendicular to the contact surface. Tension acts along a taut string, away from the body. Friction opposes relative sliding or its tendency.

Newton's first law

Zero resultant force means constant velocity (including rest).

An object has forces 30 N right, 30 N left, 40 N up and 5 N down. Find the resultant.



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1. Forces act on a particle: P N right, 3 N left, 5 N up and 5 N down. Find the resultant.
2. Forces act on another particle: 5 N right, 7 N left, 4 N up and 6 N down. Find the resultant and its magnitude.

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Forces as Vectors 5

1. Forces $(2, 3), (4, -1), (-3, 2), (a, b)$ N are in equilibrium. Find a, b.
2. With $\mathbf{i}$ east and $\mathbf{j}$ north, find the resultant, magnitude and bearing of forces $(2, 1), (3, -2), (-1, 4)$ N.

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Test Your Understanding 6

1. Forces $(7\mathbf{i} - 9\mathbf{j}), (5\mathbf{i} + 6\mathbf{j}), (p\mathbf{i} + q\mathbf{j})$ N are in equilibrium. Find p, q. Remove the third force: find the resultant's magnitude and angle with $\mathbf{j}$, to the nearest degree.
2. Forces $(\mathbf{i} - 3\mathbf{j})$ N and $(p\mathbf{i} + 2p\mathbf{j})$ N act on a particle. Find the first force's angle with $\mathbf{j}$, and $p > 0$ if the resultant is parallel to $\mathbf{i}$.

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3. Forces $(a, 2b), (-2a, -b), (3, -4)$ N act on an object in equilibrium. Find a, b.

7. Forces $(2a\mathbf{i} + 2b\mathbf{j}), (-5b\mathbf{i} + 3a\mathbf{j}), (-11\mathbf{i} - 7\mathbf{j})$ N are in equilibrium. Find a, b.

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9. Forces $\mathbf{F}_1 = (3\mathbf{i} - 2\mathbf{j})$ N and $\mathbf{F}_2 = (a\mathbf{i} + 2a\mathbf{j})$ N act on a particle, where $a > 0$.
   a. Find the angle between $\mathbf{F}_2$ and $\mathbf{i}$.
   b. Their resultant is parallel to $13\mathbf{i} + 10\mathbf{j}$. Find a.

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10. Three forces are $\mathbf{F}_1 = (-7, -4)$ N, $\mathbf{F}_2 = (4, 2)$ N and $\mathbf{F}_3 = (a, b)$ N.
    a. Given equilibrium, find a, b.
    b. Remove $\mathbf{F}_1$. Find the resultant's magnitude and angle above the horizontal, to the nearest degree.

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EXTENSION 10

Extension

A force F_1 acts in the direction of $\mathbf{i}$; a force F_2 acts at angle θ to $\mathbf{i}$. Show that the resultant has magnitude

$\sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos\theta}$.

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Newton's second law

$\mathbf{F}_{\text{resultant}} = m\mathbf{a}$

1. A 2000 kg car has driving force 800 N and resistance 200 N. Find its acceleration.
2. Find the weight of a 50 kg child.
3. A falling 70 kg body experiences 300 N air resistance. Find its acceleration.

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Combining F=ma with suvat equations 12

A 5 kg body is pulled from rest along a rough horizontal table by a horizontal force of 20 N. Friction is 4 N. Find:

a. its acceleration;
b. its displacement in the first 4 s;
c. the normal reaction.

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Testing Your Understanding 13

A particle of mass 4 kg reaches soft horizontal ground with downward speed 28 m/s. The ground exerts a constant upward resistive force of 5000 N. Find how far the particle sinks before coming to rest.

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Exercise 10C – Y1 Mech Pages 165 14

9. A lift of mass 500 kg carries passengers of total mass 300 kg. It accelerates from rest to 3 m/s over 5 m. Find:
   a. the acceleration magnitude;
   b. cable tension when accelerating downwards;
   c. cable tension when accelerating upwards.

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10. A 50 kg trolley is pulled from rest along a horizontal path by a horizontal rope. After 2 s its speed is 1 m/s. Constant resistance is 20 N. Find:
    a. its acceleration;
    b. rope tension.

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11. A 400 kg van moves at 16 m/s when its engine cuts out. It stops without braking. Model the resistance as a constant 200 N. Find:
    a. the stopping time;
    b. stopping distance;
    c. a limitation of this model.

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Motion in two dimensions

Force, acceleration, velocity and displacement are vectors. Mass, area and volume are scalars.

A resultant force $(3\mathbf{i} + 8\mathbf{j})$ N acts on a 0.5 kg particle. Take $\mathbf{i}$ east and $\mathbf{j}$ north. Find its acceleration vector, magnitude and bearing.

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Test Your Understanding 18

A 60 kg boat experiences forces $(80, 50)$, $(10p, 20q)$ and $(-75, 100)$ N. Its acceleration is $(0.8, -1.5)$ m/s^2. Find p, q.

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3. A particle of mass 3 kg has acceleration $(7\mathbf{i} - 3\mathbf{j})$ m/s^2. Find its resultant force, magnitude and bearing. Take $\mathbf{i}$ east and $\mathbf{j}$ north.

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6. Forces $(2\mathbf{i} + 3\mathbf{j})$ N and $(p\mathbf{i} + q\mathbf{j})$ N have resultant parallel to $-\mathbf{i} + 4\mathbf{j}$. Show that $4p + q + 11 = 0$.

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7. A 4 kg particle starts from rest under constant force $(6\mathbf{i} + b\mathbf{j})$ N, bearing 045°. Find:
   a. b;
   b. force magnitude;
   c. acceleration magnitude;
   d. distance travelled in 5 s.

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10. Forces $(2, 5)$ N and (p, q) N act on a particle of mass m.
    a. Their resultant is parallel to $(1, -2)$. Show $2p + q + 9 = 0$.
    b. Given $p = 1$ and acceleration magnitude $15\sqrt{5}$ m/s^2, find m.

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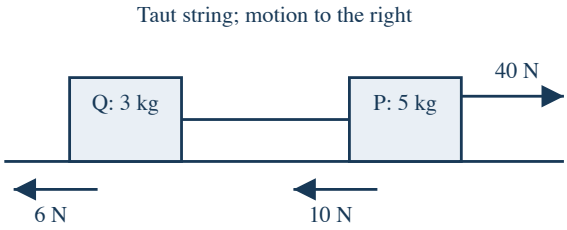
Challenge

A 0.5 kg particle experiences forces $-4\mathbf{i}$ N and $(k\mathbf{i} + 2k\mathbf{j})$ N. Its acceleration magnitude is $8\sqrt{17}$ m/s^2. Find $k > 0$.

Challenge

Particles P, Q, masses 5 kg and 3 kg, are joined by a light inextensible string on a rough horizontal plane. A horizontal 40 N force pulls P away from Q. Friction is 10 N on P and 6 N on Q.

a. Find their acceleration.



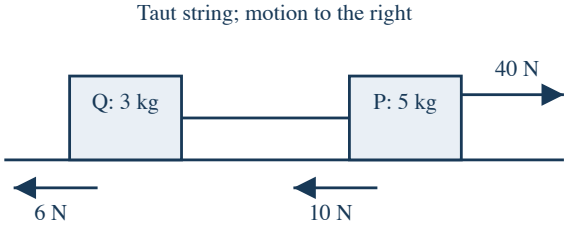
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Particles P, Q, masses 5 kg and 3 kg, are joined by a light inextensible string on a rough horizontal plane. A horizontal 40 N force pulls P away from Q. Friction is 10 N on P and 6 N on Q.

b. Find the tension.

c. Explain the string assumptions.



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Newton's third law

Interaction forces on two different bodies are equal in magnitude and opposite in direction.

A light scale-pan carries A (0.4 kg) on top of B (0.6 kg). A light vertical string raises the pan with acceleration $0.5\ \mathrm{m/s^2}$. Find:

a. string tension;

b. force on B by A;

c. force on B by the pan.

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A 50 kg woman stands in a 950 kg lift. It moves upwards with constant deceleration $2\ \mathrm{m/s^2}$. Find:

a. cable tension;

b. the force on the woman by the floor.

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1. Particles P, Q of masses 8 kg and 2 kg are joined by a light inextensible taut string on a smooth horizontal plane. A force F pulls P away from Q, giving acceleration $0.4\ \text{m/s}^2$. Find F, tension and explain the assumptions.

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5. A lorry of mass m kg tows a trailer of mass $3m$ kg using a light inextensible tow-bar. Driving force is 50,000 N; resistances are 4,000 N on the lorry and 10,000 N on the trailer. Acceleration is $5\ \text{m/s}^2$. Find both masses and tow-bar tension; explain the assumptions.

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6. Particles A, B of masses 10 kg and 5 kg are joined by a light inextensible vertical string, with B below A. A 180 N upward force acts on A. Find acceleration and tension.

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7. Particles A, B, masses 6 kg and m kg, are joined by a light inextensible vertical string, with B below A. A force of 118 N acts upwards on A, producing upward acceleration $2\ \text{m/s}^2$. Find m and the tension.

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8. A 6400 kg engine pulls a 1600 kg carriage using a light horizontal coupling. Driving force is 12,000 N; resistances are R N on the engine and 2000 N on the carriage. Acceleration is $0.5\ \text{m/s}^2$. Find R and show the coupling tension is 2800 N.

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9. A 900 kg car pulls a 300 kg trailer through a light tow-bar. Resistances are 200 N and 100 N respectively; driving force is 1200 N.
   a. Show acceleration is $0.75\ \text{m/s}^2$.
   b. Find tension.
   c. The engine force becomes zero and brakes exert F N on the car. Resistances stay the same and the bar's thrust is 100 N. Find F.

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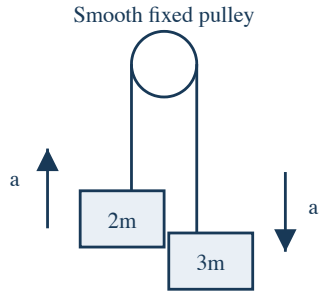
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Pulleys

Particles P, Q, masses $2m, 3m$, hang from a light inextensible string over a small smooth fixed pulley. Release from rest with string taut.

Find their acceleration, tension, force on the pulley by the string, and Q's displacement after 4 s, assuming neither particle reaches an obstruction.



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Horizontal and vertical string 34

Particles A (0.4 kg) and B (0.8 kg) are connected by a light inextensible string over a smooth fixed table-edge pulley. A is on a rough table, 4.5 m from the pulley; B hangs 0.5 m above ground. Friction on A is $0.08g$ N. Released from rest, find acceleration, time until B lands, and total distance travelled by A.

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Test Your Understanding 35

Particles A, B, masses $5m, km$ with $k < 5$, hang at equal heights from a light inextensible string over a smooth fixed pulley. Released from rest, A descends at acceleration $g/4$.

a. Show $T = 15mg/4$.

b. Find k.

c. Explain "smooth".

d. A hits ground after 1.2 s. Find B's greatest height above ground, assuming it does not reach the pulley.

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Exercise 10F – Y1 Mech Pages 175-177 36

1. Masses 4 kg and 3 kg hang from a light inextensible string over a smooth fixed pulley, released from rest.

   a. Find tension.

   b. The 4 kg mass falls 2 m before landing. Find its impact speed.

   c. Find the other mass's greatest height above its initial position, assuming it does not reach the pulley.

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2. Masses $km, 3m$ hang initially at equal heights from a light inextensible string over a smooth fixed pulley. On release the heavier mass descends at $g/3$ and hits ground after 1.8 s.
   a. Find tension.
   b. Show $k = 1.5$.
   c. Explain "smooth".
   d. Show the lighter mass reaches height $1.26g$ m above ground without reaching the pulley.

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3. Masses $A = m$ kg and $B = 3$ kg, with $m > 3$, hang from a light inextensible string over a smooth fixed pulley. Released from rest, A descends 2.5 m in 1.25 s and stops on the ground.
   a. Show acceleration is $3.2\ \text{m/s}^2$.
   b. Find tension.
   c. Show $m = 65/11$.

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3. Continued: A hits the ground after descending for 1.25 s with acceleration $3.2\ \mathrm{m/s^2}$. The string becomes slack and B moves freely without reaching the pulley.
   d. Explain how inextensibility was used.
   e. Find the time from impact until the string becomes taut again.

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4. A 5 kg particle on a rough horizontal table is connected to a hanging 3 kg particle by a light inextensible string over a smooth fixed pulley. Friction is 24.5 N. Released from rest, find acceleration and tension.

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5. A 2.5 kg box on a rough table is connected to a hanging 1.5 kg sphere by a light inextensible string over a smooth fixed pulley. Friction is k N. Released from rest, the sphere falls 0.8 m in 0.75 s.
   a. Find acceleration, tension and k.
   b. Explain how inextensibility was used.

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